

Phase composition and solid solution strengthening effect in TiZrNbMoV high-entropy alloys



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ABSTRACT

TiZrNbMo_xV_y high-entropy alloys (HEAs) with $x = 0-2$, $y = 1$ and $y = 0.3$, respectively, were designed and prepared by copper mold casting technology. The phase composition and stability of these HEAs were investigated. It is shown that the HEAs with low content of V are composed of only one type of bcc solid solution phase (SSP), and demonstrate excellent phase stability at 1273 K. The high content of V and Mo results in the formation of two types of bcc SSPs and the decrease of phase stability in the HEAs. Based on the previously proposed criteria, the formation ability of solid solution phase for this kind of HEAs was comprehensively evaluated. The compressive mechanical properties of the as-cast and annealed HEAs were measured. It has been found that Mo plays a strong solid solution strengthening effect on this kind of HEAs. Especially, TiZrNbMo_{0.3}V_{0.3} has the yield strength and plastic strain of 1312 MPa and >50%, respectively, and still maintains the excellent plastic deformation ability even after annealed at 1273 K for 72 h. The strengthening effect in this kind of HEAs is considered to be due to the shear modulus mismatch. The solubility limit of HEAs is correspondent to shear modulus mismatch of 29.

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1. Introduction

Historically, metallic materials have been principally designed in the way that an alloy has one or two kinds of predominant elements, and all the other elements in this alloy only play an ancillary role. In recent years, this dogma in the composition design of metallic materials has been broken by a subversive concept named as “high-entropy”. The high-entropy alloys (HEAs) were designed according to the principle of equal-molar or near-equalmolar fraction for all the components, which then results in a high configurational entropy in the alloy [1,2]. The high-entropy is beneficial to the stabilization of disordered solid solution phases such as a body- or face-centered cubic (fcc), and disadvantageous to the formation of ordered intermetallic crystalline phases. Nevertheless, it is noticed that the configurational entropy only takes into account of a part of the entire entropy for a certain alloy. This means that only a few of, but not all the alloy system can form single solid solution phase such as face-centered cubic (fcc) [3–5], body-centered cubic (bcc) [6–9], and orthorhombic [10] phase.

HEAs have attracted great attraction due to their high strength and ductility, excellent oxidation resistance and thermal stability, etc. [2,11,12,37–40]. Of the HEAs discovered to date, bcc type of

HEAs exhibit enhanced strength and excellent phase stability at high temperature. For instance, the tensile strengths of FeCoCrNiMn HEAs are under 600 MPa [13,14] when they have fcc type of structure. As Al element is added into FeCoCrNiMn HEAs, the tensile strength can be increased up to 1150 MPa [14] due to the formation of bcc type of crystalline structure. For high temperature applications, Senkov et al. developed refractory NbMoTaWV and TaNbHfZrTi HEAs [6,7], which have bcc type of structure and demonstrate high temperature strength and excellent phase stability at 1473 K. However, it is seen that these HEAs still have the deficiencies of high cost and density due to the existence of Hf and Ta elements. In order to decrease the density and further improve the oxidation resistance of refractory HEA, Senkov et al. investigated the CrNbTiVZr HEAs [36]. They found that the Cr bearing HEAs contain a lot of Laves phases and exhibit great brittleness at room temperature. Therefore, it is significant to explore new refractory HEAs with low cost, low density and high ductility. Alternatively, TiZrNbMoV HEAs are promising for high temperature engineering and manufacture industries considering that they are cheaper and lighter than NbMoTaW HEAs. As for TiZrNbMoV HEAs, Tian et al. studied the equilibrium bulk properties of TiZrNbMoV HEAs by using *ab initio* method [15]. Zhang et al. investigated the effect of V on the phase formation and mechanical properties [16]. It has been shown that the alloying effect of V element in this kind of HEAs can be negligible.

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Mo is one of the most important alloying elements in high temperature Ti-based alloys [17,18]. It is reasonable to deduce that Mo may also play a critical role in this kind of Ti bearing HEAs because Mo has highest melting point among all the components of TiZrNbMoV alloy. However, Laves phases may be also formed in this kind of HEAs if excessive Mo is added. Therefore, it is necessary to investigate the effects of Mo on the phase evolution and mechanical properties of this kind of HEAs. It is also noticed that most bcc HEAs discovered to date usually exhibit insufficient plasticity, which may be unfavorable to the practical engineering application. The shortage of poor plastic deformation for bcc HEAs is not only due to the crystalline structure but also affected by the properties of the components. Therefore, it is highly needed to optimize the compositions of this kind of HEAs so as to achieve high comprehensive properties.

In this work, the microstructural evolution and phase composition of TiZrNbMo_xV_y HEAs with $x = 0-2$, while $y = 1$ and $y = 0.3$, respectively are investigated. The phase stability of these HEAs is also explored at 1273 K to provide a reference for the high temperature application. The room temperature compressive properties are studied, and the optimal compositions which have fine comprehensive mechanical properties are hopefully to be pin-pointed out. Finally, the strengthening mechanism for this kind of HEAs is discussed in detail.

2. Experimental details

Two groups of TiZrNbMo_xV_y HEAs were designed and investigated in this work. One group of HEAs is TiZrNbMo_xV (denoted

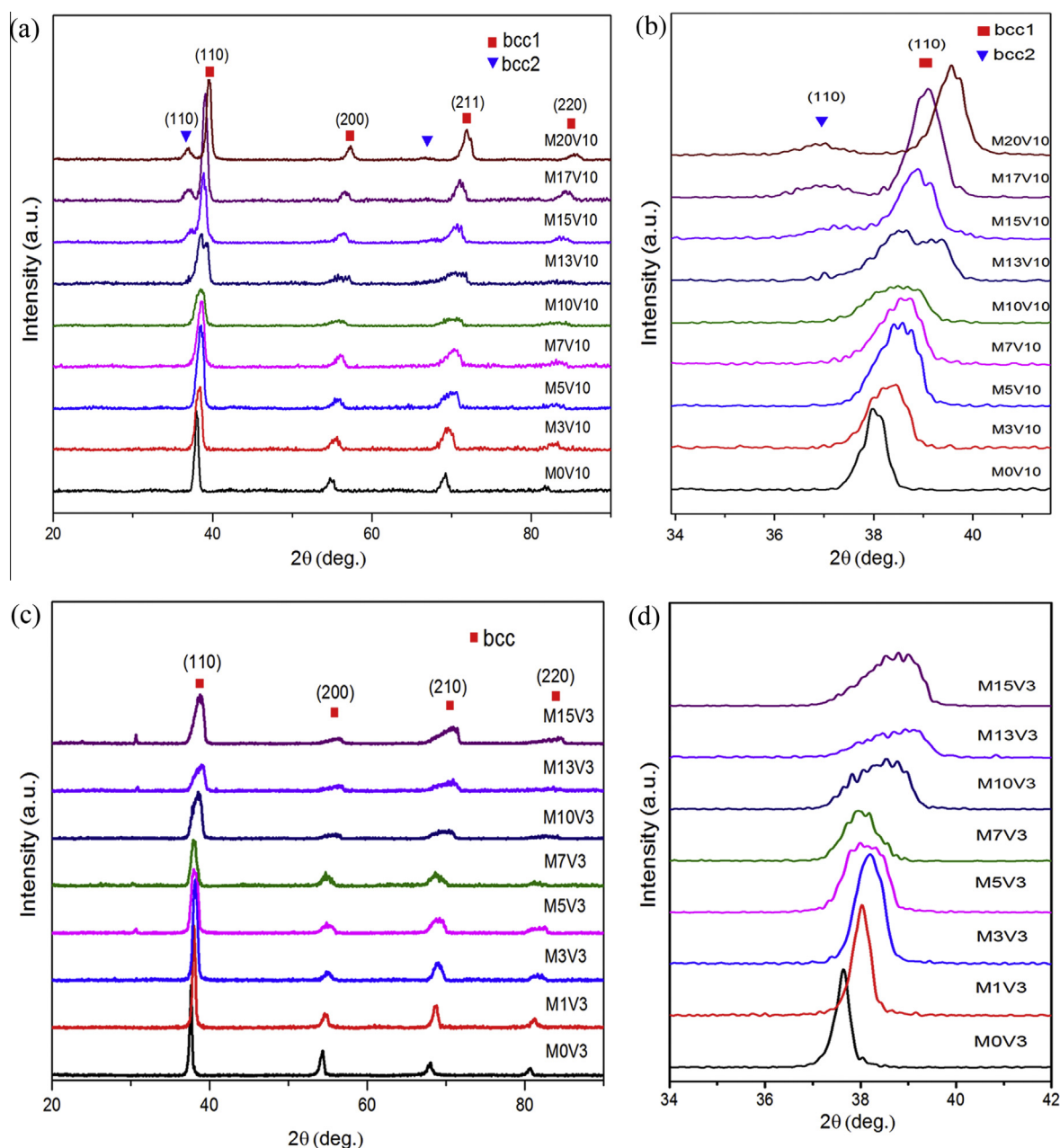


Fig. 1. (a) and (c) The XRD patterns of G-MV10 and G-MV3 HEAs, (b) and (d) the magnified (110) peaks as shown in (a) and (c), respectively.

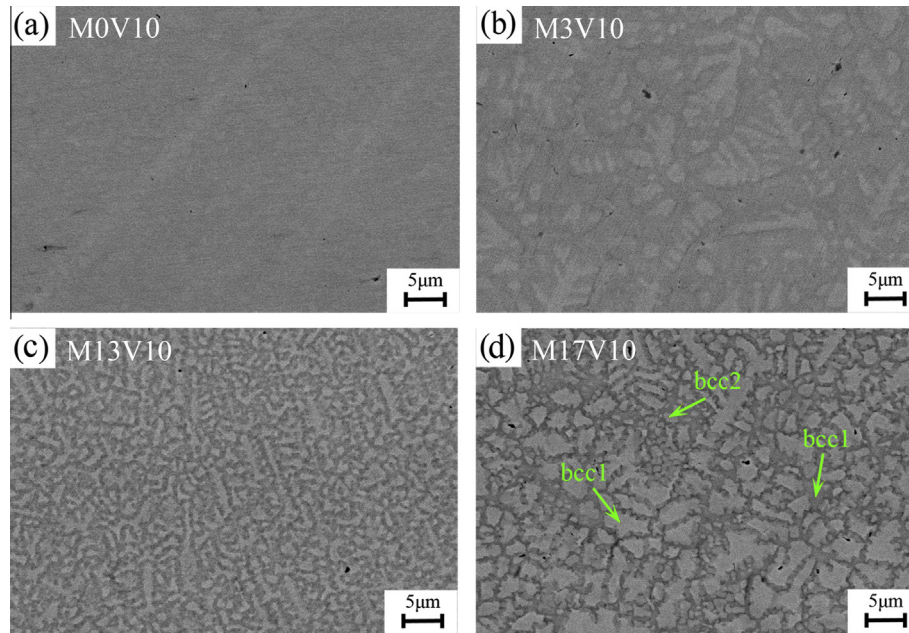


Fig. 2. The SEM backscattering electron micrographs of as-cast (a) M0V10, (b) M3V10, (c) M13V10 and (d) M17V10 alloy.

Table 1

The chemical composition in different region of TiZrNbVMo HEAs measured by EDS.

Alloys	Region ^a	Chemical compositions (at.%)				
		Ti	Zr	Nb	V	Mo
M0V10	Nominal	25	25	25	25	–
	O	26.30 ± 0.19	25.92 ± 0.15	21.40 ± 0.13	26.38 ± 0.20	–
M3V10	Nominal	23.26	23.26	23.26	23.26	6.97
	DR	23.75 ± 0.22	16.69 ± 0.14	27.86 ± 0.19	22.61 ± 0.22	9.09 ± 0.13
	IR	23.79 ± 0.19	23.62 ± 0.15	21.97 ± 0.16	24.30 ± 0.20	6.32 ± 0.11
M13V10	Nominal	18.87	18.87	18.87	18.87	24.52
	DR	17.99 ± 0.21	15.97 ± 0.15	22.46 ± 0.19	19.16 ± 0.22	24.42 ± 0.20
	IR	18.52 ± 0.20	20.17 ± 0.16	20.17 ± 0.18	19.06 ± 0.21	22.08 ± 0.18
M17V10	Nominal	17.54	17.54	17.55	17.55	29.82
	DR	16.07 ± 0.22	6.19 ± 0.12	19.16 ± 0.19	15.43 ± 0.22	43.15 ± 0.26
	IR	18.23 ± 0.22	24.99 ± 0.19	13.36 ± 0.18	16.02 ± 0.22	27.41 ± 0.22
	PT	16.89 ± 0.25	42.82 ± 0.29	8.58 ± 0.20	15.65 ± 0.25	16.06 ± 0.22
M0V3	Nominal	30.3	30.3	30.3	9.09	–
	O	31.09 ± 0.23	31.37 ± 0.18	28.26 ± 0.17	9.28 ± 0.15	–
M3V3	Nominal	27.77	27.77	27.77	8.34	8.34
	O	28.72 ± 0.21	30.15 ± 0.17	24.39 ± 0.16	9.01 ± 0.14	7.74 ± 0.12
M5V3	Nominal	26.32	26.32	26.32	7.90	13.16
	DR	23.42 ± 0.15	16.71 ± 0.11	37.60 ± 0.17	5.43 ± 0.10	16.84 ± 0.12
	IR	25.74 ± 0.18	37.63 ± 0.17	18.37 ± 0.13	9.29 ± 0.12	8.98 ± 0.11
M7V3	Nominal	25	25	25	7.5	17.5
	DR	27.61 ± 0.16	19.71 ± 0.11	26.04 ± 0.14	6.24 ± 0.10	20.41 ± 0.12
	ID	27.22 ± 0.16	31.72 ± 0.14	18.16 ± 0.12	9.04 ± 0.11	13.87 ± 0.11
M7V3-AN	DR	26.77 ± 0.22	16.84 ± 0.13	28.89 ± 0.18	8.84 ± 0.15	18.67 ± 0.16
	IR	25.16 ± 0.20	36.87 ± 0.19	21.48 ± 0.17	5.47 ± 0.12	11.03 ± 0.14
	PT	7.33 ± 0.08	36.15 ± 0.30	9.58 ± 0.33	15.78 ± 0.17	31.16 ± 0.36
M10V3	Nominal	23.25	23.25	23.25	6.98	23.25
	DR	18.13 ± 0.11	12.74 ± 0.14	28.33 ± 0.31	4.96 ± 0.08	35.84 ± 0.32
	IR	20.59 ± 0.11	21.82 ± 0.29	22.83 ± 0.35	6.62 ± 0.09	28.15 ± 0.34
M13V3	Nominal	21.74	21.74	21.74	6.52	28.26
	DR	16.74 ± 0.18	10.25 ± 0.13	25.53 ± 0.30	4.28 ± 0.16	43.20 ± 0.33
	IR	18.60 ± 0.10	47.90 ± 0.31	9.10 ± 0.31	7.20 ± 0.08	17.20 ± 0.33

^a Nominal: nominal composition, O: overall, DR: dendrite, IR: interdendrite, PT: precipitation regions.

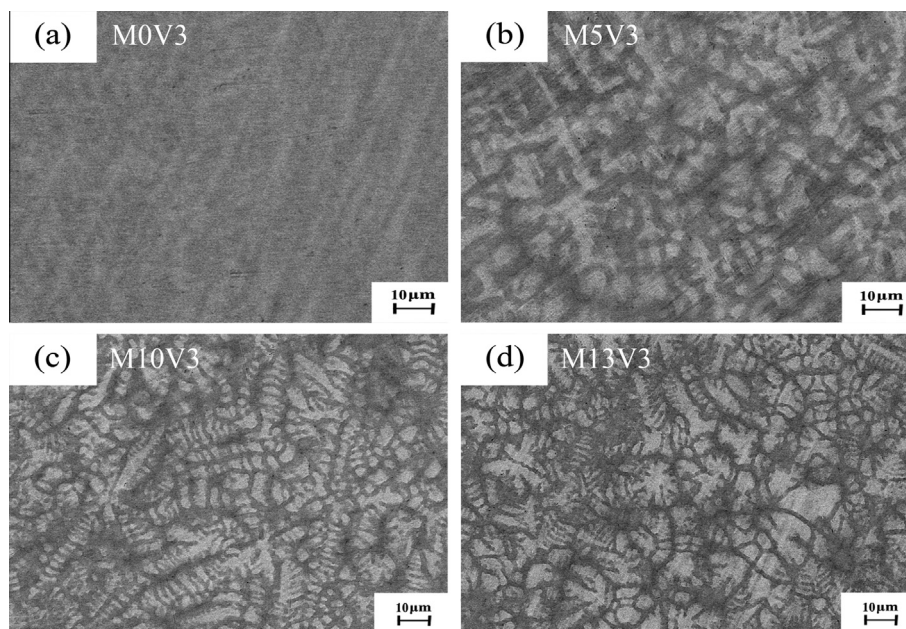


Fig. 3. The SEM backscattering electron micrographs of as-cast (a) M0V3, (b) M5V3, (c) M10V3 and (d) M13V3 alloy.

as G-MV10), where $x = 0, 0.3, 0.5, 0.7, 1.0, 1.3, 1.5, 1.7$ and 2.0 (denoted as M0V10, M3V10, M5V10, M7V10, M10V10, M13V10, M15V10, M17V10 and M20V10 in the following context), respectively. The other is $\text{TiZrNbMo}_x\text{V}_{0.3}$ (denoted as G-MV3), where $x = 0, 0.1, 0.3, 0.5, 0.7, 1.0, 1.3$ and 1.5 (denoted as M0V3, M1V3, M3V3, M5V3, M7V3, M10V3, M13V3 and M15V3), respectively. The alloy ingots were prepared by arc-melting a mixture of pure metals (with the purity >99 wt.%) in a water-cooled copper hearth under a Ti-gettered argon atmosphere. To ensure the chemical homogeneity, the alloys were flipped and remelted five times. The melted alloys were then drop-cast into a water-cooled copper mold to obtain cylindrical rods with a diameter of 3 mm.

The phases of the interested alloys were analyzed by X-ray diffraction (XRD) using a PHILIPS APD-10 diffractometer (Philips, Amsterdam, the Netherlands) with $\text{Cu K}\alpha$ radiation. The microstructure of the samples was investigated by using ZEISS SUPRA 55 scanning electron microscope (SEM) with energy-dispersive spectrometry (EDS). To investigate the microstructural stability of the HEAs, ingots with 60 g were sealed in quartz tube after evacuated and refilled with high purity argon. Then these tubes were annealed at 1273 K for 72 h. The room temperature compressive tests were conducted by using an MTS 809 materials testing machine and cylindrical rods of $\phi 3 \text{ mm} \times 4.5 \text{ mm}$ according to the standard of ASTM: E9-09. The strain rate for the compressive test is $2 \times 10^{-4} \text{ s}^{-1}$.

3. Results and discussions

3.1. Microstructural evolution

The X-ray diffraction patterns of TiZrNbVMo alloys are shown in Fig. 1. Two types of bcc structure can be differentiated in G-MV10 HEAs. In this group of HEAs with low content of Mo (M0V10–M10V10), only a single bcc structure is formed. As the amount of Mo is increased to $x = 1.3$ (M13V10), a new minor (110) peak emerges near the original (110) peak of the major bcc structure. To shed light on the effect of Mo on the structure of HEAs, the (110) peaks for as-cast alloys are magnified, as shown in Fig. 1(b). It is seen that for the HEAs with low content of Mo

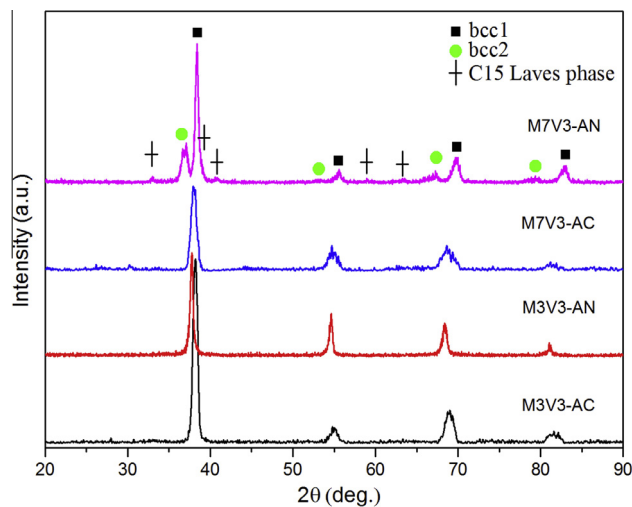


Fig. 4. The XRD patterns of M3V3 and M7V3 in as-cast state and after annealed at 1273 K for 72 h.

(M0V10–M10V10), the (110) peak shifts towards high angle side as Mo content is increased. The reduction of lattice constant is mainly owing to the solid solution of relatively small radius of Mo atoms. It is also found that with the addition of Mo, the (110) peak of the original bcc phase is apparently broadened, meaning that there are severe distortion in the lattice structure and a trend of phase decomposition. As the content of Mo is increased to $x = 1.5$ (M13V15–M20V10), the original (110) peak of HEAs is found to split into two sub-peaks. A weak peak arises when $x \geq 1.5$ at the 2θ around 37° , indicating the formation of a new bcc phase. The original peak keeps on shifting towards higher angle side, while the new peak shifts towards lower angle side. The XRD results show that the new bcc phase is rich of the components with relative larger radii among the alloy system.

The alloying effect of Mo on the phase structure of G-MV3 HEAs is shown in Fig. 1(c) and (d). It is seen that only a single phase with bcc crystal structure is formed in this group of HEAs. Fig. 1(d)

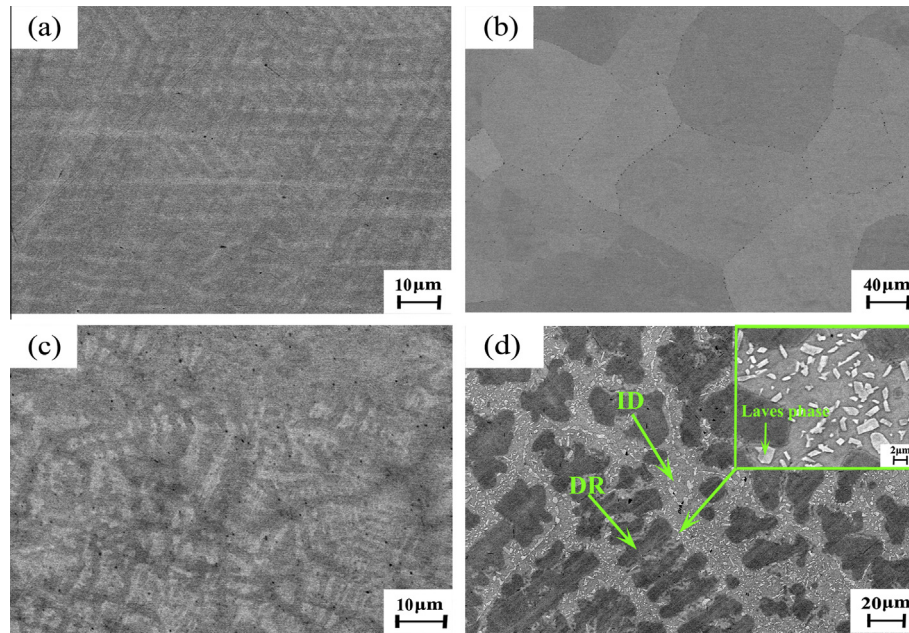


Fig. 5. The SEM back scattering electron micrographs of M3V3 and M7V3 HEAs, (a) as-cast M3V3, (b) annealed M3V3, (c) as-cast M3V3, and (d) annealed M7V3 at 1273 K for 72 h. The inset in (d) is the magnified image of interdendrite region revealing the precipitation of Laves phase.

indicates that as the concentration of Mo is increased, the (110) peak is broadened, and shifts towards the higher angle side. This result illustrates that all the components are solved into the bcc phase. Severe distortion in the crystal lattice emerges due to the mixing of these five kinds of components with different atomic radii.

To further investigate the phase formation in this alloy system, the SEM backscattering electron micrographs of as-cast G-MV10 HEAs with different amount of Mo are presented in Fig. 2. It is shown that M0V10 alloy has almost uniform microstructure. Dendrite structure is observed in M3V10 due to the addition of Mo. The dendrites in these HEAs become thinner and denser with the increase of the amount of Mo, which also reflects the segregation of solutes in the interdendrites. It is seen that this kind of segregation is caused by the enrichment of Zr, which is deduced by the EDX results as shown in Table 1. It is seen that the content of Zr in dendrites is lower than the nominal composition, and the distributions of Mo however are opposite. In M17V10 HEA (Fig. 2(d)), a new phase is formed in the interdendritic region. It is shown from Table 1 that the contents of Zr and Mo in the dendrite of M17V10 HEA are 6.19% and 43.15%, deviating obviously from their nominal composition. The interdendritic region is slightly Nb-depleted. It is also be found that the new bcc phase is enriched with Zr. The content of Zr in this new phase is as high as 42.82%.

The microstructural evolution of G-MV3 HEAs is shown in Fig. 3. It is seen that this group of HEAs has typical single solid solution phase with different extent of heterogeneity. The SEM backscattering electron micrograph of M0V3 HEA exhibits weak contrast, meaning that this alloy has fine homogeneous microstructure. As shown in Table 1, the compositional distribution rules of all the components in this group are same as those in G-MV10 group. The enrichment of Zr in the interdendrites is intensified with the increase of Mo. From the EDX results listed in Table 1, it is seen that the content of V in the interdendrite region is still higher than the nominal composition. One of the reason for the suppression of second phase is due to the relative low content of V. In other words, the formation of second bcc phase in G-MV10 HEAs is attributed to the unified effect of high content of V and Mo.

Table 2

Relevant parameters of proposed criteria for TiZrNbMoV alloys (δ was amplified by 100 for clarity).

	δ	ΔH_{mix} (kJ/mol)	ΔS_{mix} (J/mol/K)	Ω	$\Delta\chi$ (%)	G_a (GPa)	ΔG
M0V10	5.43	−0.25	11.53	103.76	5.42	41.5	13.35
M3V10	5.51	−1.26	12.83	23.46	5.33	40.0	18.89
M5V10	5.54	−1.78	13.15	15.86	5.26	39.1	21.50
M7V10	5.55	−2.21	13.31	14.14	5.20	38.3	23.61
M10V10	5.55	−2.72	13.38	11.71	5.10	37.2	26.16
M13V10	5.54	−3.10	13.33	10.35	5.00	36.2	28.17
M15V10	5.53	−3.31	13.25	9.73	4.94	35.6	29.30
M17V10	5.51	−3.47	13.15	9.26	4.88	35.7	30.29
M20V10	5.47	−3.67	12.98	8.73	4.79	34.3	31.56
M0V3	4.72	1.43	10.83	17.12	4.20	39.5	12.68
M1V3	4.84	0.80	11.62	33.34	4.26	38.9	15.19
M3V3	5.03	−0.28	12.32	102.68	4.35	37.9	18.93
M5V3	5.17	−1.14	12.65	26.14	4.40	36.9	21.71
M7V3	5.27	−1.83	12.80	16.65	4.43	36.1	23.89
M10V3	5.37	−2.62	12.83	11.81	4.45	35.0	26.45
M13V3	5.42	−3.20	12.72	9.73	4.44	34.0	28.41
M15V3	5.44	−3.49	12.61	8.90	4.42	33.4	29.48

3.2. The phase stability

The HEAs in present work were prepared by arc melting in water-cooled copper mold, resulting in non-equilibrium solidification microstructure. During the non-equilibrium solidification process, the redistribution of solutes between bcc structure and liquid is retarded and the crystallization of second phase is hindered. Therefore, it is necessary to investigate the phase stability and mechanical properties of these HEAs for their engineering application.

Fig. 4 reveals the XRD patterns of as-cast and annealed M3V3 and M7V3 HEAs. It is shown that after annealed at 1273 K for 72 h, M3V3 still preserve the single bcc solid solution phase, while M7V3 has been decomposed into two bcc phases and C15 Laves phase. The diffraction peaks in the XRD pattern of M3V3 alloy become sharper after annealing, which may be caused by the decrease of inner stress and the coalescence of crystalline grains. The phase stability can be

also investigated by the SEM images of HEAs. As shown in Fig. 5, the as-cast M3V3 has single phase microstructure with trivial interdendritic segregation. After homogenized at 1273 K for 72 h, the microstructure is still composed of one single phase, and only the crystalline grains have been changed to be equiaxed grain structure. The microstructure experienced recovery, homogenization and grain growth so that straight grain boundaries have been formed. As for M7V3 alloy, the as-cast microstructure is composed of single

bcc phase. After annealing at 1273 K for 72 h, a new bcc and a Laves phase were formed in the interdendrite region. From the magnified section shown in Fig. 5(d), the Laves phase was actually precipitated from the new bcc phase. It can be imaged from the SEM results that the sequence of microstructural evolution for M7V3 HEA during the annealing process is as following: the new bcc phase is precipitated in the interdendrites and then the Laves phase is formed from the new bcc phase.

The EDS results listed in Table 1 reveal that the dendrite region of as-annealed M7V3 is slightly Zr-depleted, indicating the bcc phase with smaller lattice constant. And the interdendrite region is Zr-enriched, which is correspondent to the new bcc phase with larger lattice constant. It is also shown that the precipitated Laves phase is enriched with Zr, V and Mo, and extremely depleted with Ti and Nb. Combining the EDS results with XRD patterns, it can be deduced that the Laves phases are formed due to the breaking of solution limit of Zr, V and Mo in the new bcc phase, and have Mo_2Zr or V_2Zr type of structure.

3.3. Phase selection in TiZrNbMoV HEAs

Up until now, several parameters have been proposed to predict the structural stability and phase formation of HEAs. Yang and Zhang [19] proposed that the enthalpy of mixing, ΔH_{mix} , entropy of mixing, ΔS_{mix} , and the radius mismatch between elements, δ can be used as the criteria. They defined the criteria for the formation of random solid solution in HEAs to be in the range of $-15 < \Delta H_{\text{mix}} < 5 \text{ kJ/mol}$ and $1\% < \delta < 6.6\%$ [20]. It is known that at a given temperature, a solid solution phase is stable when its free energy is $\Delta G \leq 0$. This situation can be achieved either as $\Delta H_{\text{mix}} \leq 0$, or $\Delta H_{\text{mix}} \leq 0$ with the $-\Delta S_{\text{mix}}$ having higher absolute value. In Ref. [19], Zhang et al. proposed a parameter named $\Omega = T_m \Delta S_{\text{mix}} / |\Delta H_{\text{mix}}|$, where T_m is the average melting temperature for all components for evaluating the probability forming a solid solution or intermetallics phase in multicomponent systems. They found that a high value of Ω ($\Omega > 1.1$) and low value of δ ($\delta < 6.6$) are beneficial to the formation of solid solutions in multi-component alloys. For the phase composition, Poletti and Battezzati [21] investigated the electrochemical effect in the HEA systems by evaluating the difference in electronegativity, $\Delta\chi$. Instead of the empirical Pauling scale, Poletti et al. employed the electronegativity of d -electron elements to calculate $\Delta\chi$ [22]. They found that as $1\% < \delta < 6\%$ and $3 < \Delta\chi < 6$, only solid solutions are formed.

For the current alloys, relevant parameters of proposed criteria for TiZrNbMoV alloys are listed in Table 2, and the $\Delta H_{\text{mix}}-\delta$, $\Omega-\delta$ and $\Delta\chi-\delta$ graphs are shown in Fig. 6. It is evident that on the whole, all current alloys fall in the solid solution zone according to the proposed criteria. However, it should be noted that in $\Delta H_{\text{mix}}-\delta$, $\Omega-\delta$ and $\Delta\chi-\delta$ diagram, Zhang et al. also showed a region where solid-solution + intermetallics can be formed, as indicated by the green lines. Therefore, it is seen that all the G-MV10 alloys fall in the solid-solution + intermetallics region if evaluated by $\Delta H_{\text{mix}}-\delta$. This criterion is obviously ineffective for predicting the formation of solid solution in current HEAs. When the $\Omega-\delta$ criterion is employed, small amounts of HEAs which contain high content of Mo fall in solid-solution + intermetallics region overlap. Therefore, it is seen that $\Omega-\delta$ criterion is more effective in predicting phase formation of current alloys. From Fig. 6(c), the $\Delta\chi-\delta$ criterion reflects the phase composition of the current HEAs reasonably. Another finding from Fig. 6 is that the G-MV3 HEAs are more liable to form solid solution microstructure than G-MV10 HEAs since G-MV3 HEAs have lower radius mismatch and electronegativity difference.

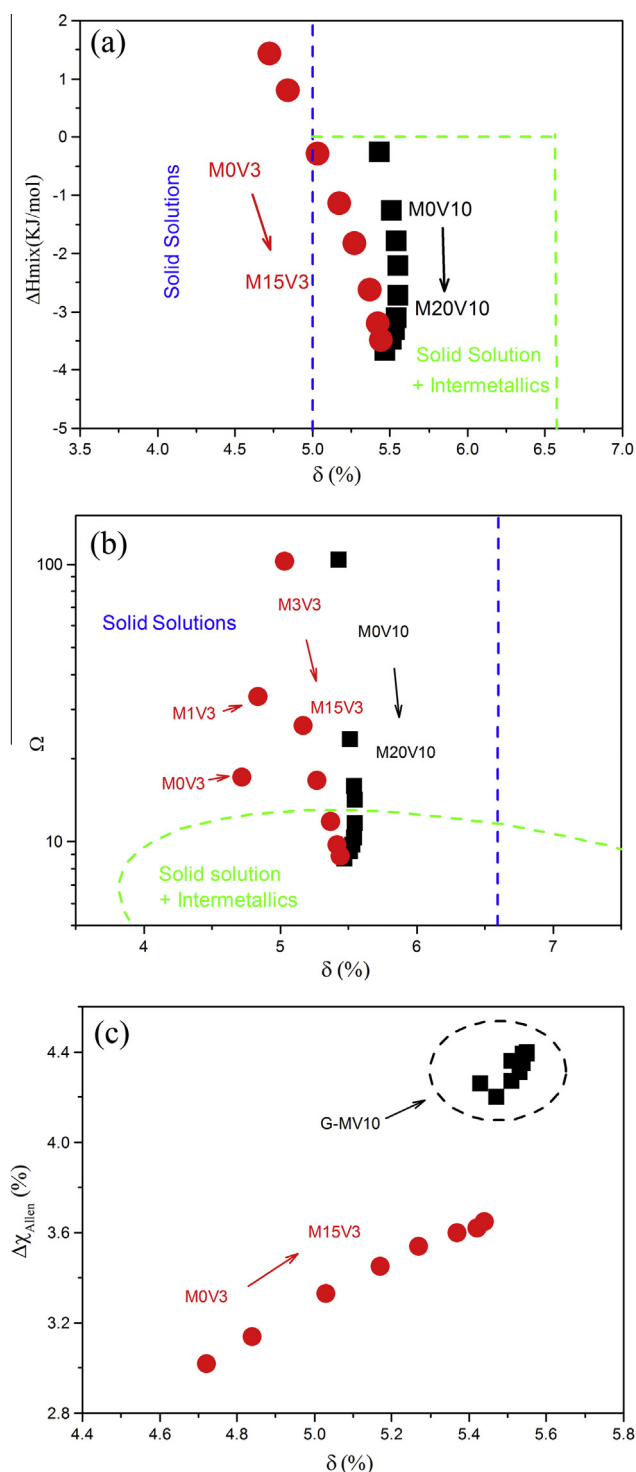


Fig. 6. The (a) $\Delta H_{\text{mix}}-\delta$, (b) $\Omega-\delta$ and (c) $\Delta\chi_{\text{Allen}}-\delta$ for TiZrNbMoV HEAs.

3.4. The compressive properties of TiZrNbVMo HEAs

The engineering stress–strain curves, yield strength and plastic strain for the present HEAs are depicted in Fig. 7. Room temperature mechanical properties of HEAs studied in present work and reported in literature are summarized in Table 3. For G-MV10 HEAs, the yield strengths increase with the addition of Mo from M0V10 to M10V10, and then decrease sharply at $x = 1.3$, which might be due to the formation of a new bcc phase. As the content of Mo keeps on rising, the yield strengths of HEAs increase again, but cannot exceed the highest value tested in M10V10 HEA. It is seen that for the HEAs with single bcc phase, the high content of Mo in these alloys results in the strengthening effect. Correspondingly, the plastic strain decreases with the increase of Mo. One of the reasons for the strengthening effect of Mo is related to the higher bulk modulus of Mo element than other components. It is seen that the plastic strains of the HEAs with single phase are above 30%, which are considerably high among bcc HEAs. The plastic strain decreased dramatically as the second bcc phase emerges. From the compressive tests, one can conclude that in G-MV10 HEAs, the low concentration of Mo results in high strength and large plastic strain. G-MV10 HEAs with the addition of Mo of $x = 0.5$ –1.0 possess excellent comprehensive mechanical properties. The mechanical properties tend to decrease as the concentration of Mo is above 20%.

The mechanical properties of G-MV3 HEAs show some different merits from those of G-MV10 HEAs. It is seen that the addition of Mo in G-MV3 HEAs increases the yield strength and decreases the plastic strain. M3V3 alloy exhibits a good combination of yield strength (~ 1312 MPa) with plastic strain ($>50\%$). This feature in

mechanical properties can be explained by the microstructure and phase composition. According to XRD analysis, there is only one type of bcc phase in G-MV3 HEAs. The more the Mo content, the higher the yielding strength due to the higher bulk modulus of Mo. From Fig. 4, it is seen that high concentration of V in G-MV10 HEAs indeed plays intensive strengthening role. Low concentration of V is beneficial to the plastic deformability.

The compressive properties of annealed M3V3 and M7V3 HEAs are shown in Fig. 8. It is seen that both the yield strengths of M3V3 and M7V3 were decreased after heat treatment as compared with those of as-cast alloys. The reduction of yielding strength of M3V3 after annealing should be attributed to the coarsening of crystalline grains and the formation of equiaxed grains as shown in Fig. 5(b). As for M7V3, the decrease in yield strength is caused by the formation of new bcc structure and the precipitation of Laves phases. It is also seen that both the plastic strains of annealed M3V3 and M7V3 are equivalent to those of as-cast alloys, indicating that the plasticity of this alloy system is mainly controlled by the bcc phase.

3.5. The solid solution strengthening effect of HEAs

As engineering materials, the specific yield strength (SYSs) of HEAs is a very important parameter for the design of a component. The SYSs of HEAs are related to their density. In this paper, we calculate the theoretical density, ρ_{mix} , of a disordered solid solution by $\rho_{\text{mix}} = \frac{\sum c_i A_i}{\sum c_i A_i / \rho_i}$, where A_i and ρ_i are the atomic weight and density of element i , respectively. The ρ_i values of alloying elements were taken from Ref. [23]. The SYSs of current alloys and other reported HEAs are obtained by using the calculated densities and measured

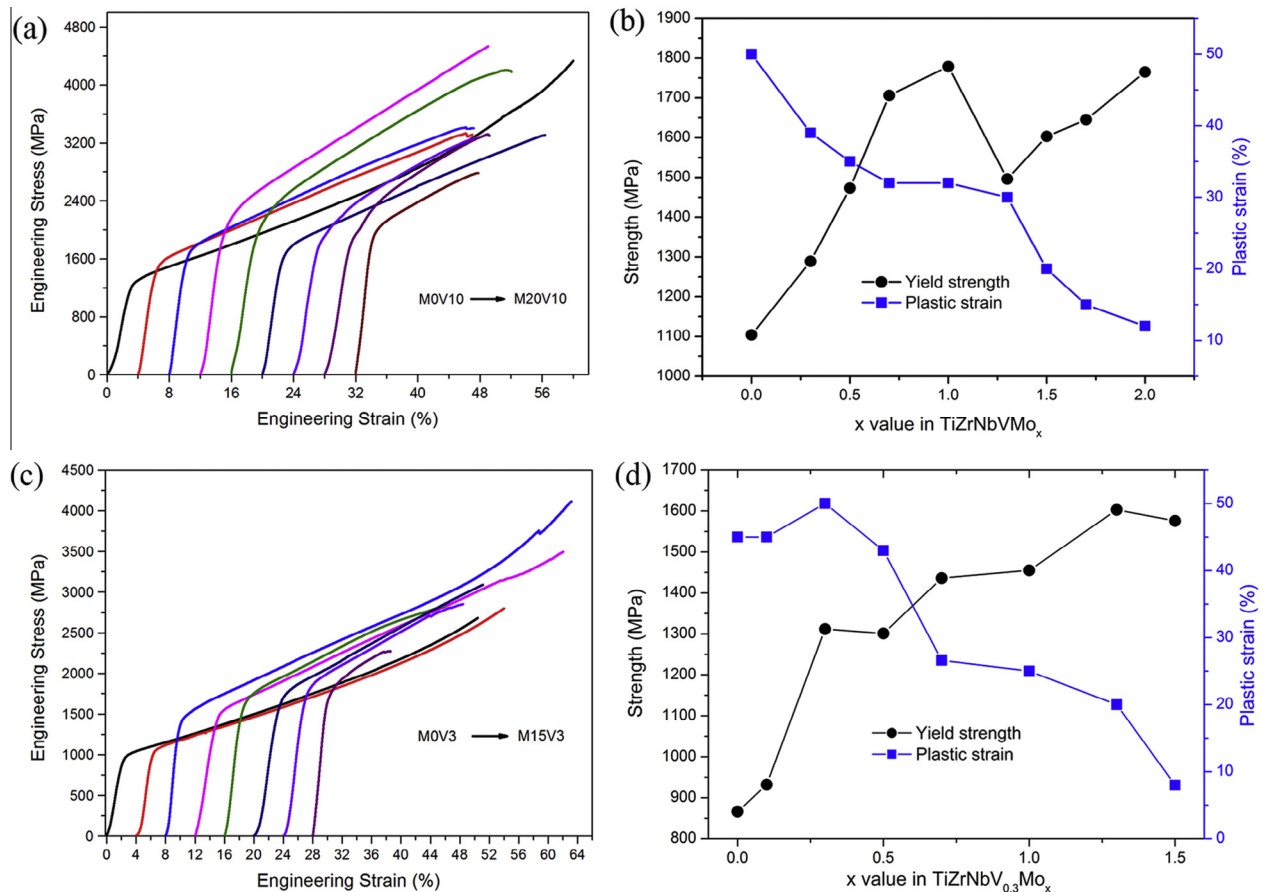


Fig. 7. The mechanical properties of as-cast G-MV10 and G-MV3 alloys at room temperature: (a) and (c) the engineering stress–strain curves of G-MV10 and G-MV3 HEAs, respectively, (b) and (d) the yield strength and plastic strain as a function of Mo concentration in G-MV10 and G-MV3 HEAs, respectively.

Table 3

Room temperature mechanical properties of HEAs studied in present work and reported in literature.

Alloy	ρ_{mix} (g/cm ³)	$\sigma_{0.2}$ (MPa)	$\sigma_{0.2}/\rho$ (kPa m ³ /kg)	ε_p (%)	Ref.
TiZrNbV	6.46	1104	171	50	Present work
TiZrNbVMo _{0.3}	6.69	1289	193	42	Present work
TiZrNbVMo _{0.5}	6.83	1473	216	32	Present work
TiZrNbVMo _{0.7}	6.96	1706	245	32	Present work
TiZrNbVMo _{1.0}	7.13	1779	250	32	Present work
TiZrNbVMo _{1.3}	7.29	1496	205	30	Present work
TiZrNbVMo _{1.5}	7.38	1603	217	20	Present work
TiZrNbVMo _{1.7}	7.48	1645	220	15	Present work
TiZrNbVMo _{2.0}	7.60	1765	232	12	Present work
TiZrNbV _{0.3}	6.51	866	133	45	Present work
TiZrNbV _{0.3} Mo _{0.1}	6.60	932	141	45	Present work
TiZrNbV _{0.3} Mo _{0.3}	6.77	1312	194	50	Present work
TiZrNbV _{0.3} Mo _{0.5}	6.93	1301	188	43	Present work
TiZrNbV _{0.3} Mo _{0.7}	7.07	1436	203	26.6	Present work
TiZrNbV _{0.3} Mo _{1.0}	7.26	1455	200	25	Present work
TiZrNbV _{0.3} Mo _{1.3}	7.43	1603	216	20	Present work
TiZrNbV _{0.3} Mo _{1.5}	7.53	1576	209	8	Present work
AlCoCrFeNiTi _{0.5}	6.44	2260	351	23.3	[26]
AlCrFeCoNi	6.72	1251	186	32.7	[27]
CrFeCoNiCuTi _{0.8}	7.58	1042	137	2.1	[28]
CrFeCoNiCuTi _{0.5}	7.82	700	90	21.6	[28]
CrFeCoNiCu	8.33	230	28	50	[28]
AlCrFeCoNiCu	7.07	1303	184	18	[29]
AlCrFeCoNiCuV	6.92	1469	212	6	[29]
Cu _{0.25} AlCoCrFeNiTi _{0.5}	6.54	1994	305	9.1	[30]
AlCoCrFeNiMo _{0.1}	6.45	1804	280	7.5	[31]
AlCoCrFeNiSi _{0.2}	6.92	1265	183	13.8	[32]
AlCoCrFeNiSi _{0.4}	6.22	1481	238	13.4	[32]
TaNbHfZrTi	9.90	929	94	50	[33]
NbMoTaW	13.64	1058	78	2.6	[6]
VNbMoTaW	12.36	1246	101	1.7	[6]
NbCrMo _{0.5} Ta _{0.5} TiZr	8.00	1595	199	5	[34]
CrNbTiZr	6.64	1260	190	6	[36]
CrNbTiVZr	6.56	1289	198	3	[36]
AlMo _{0.5} NbTa _{0.5} TiZr	7.30	2000	274	9	[35]
AlNb _{1.5} Ta _{0.5} Ti _{1.5} Zr _{0.5}	6.80	1280	188	2.5	[35]
Al _{0.4} Hf _{0.6} NbTaTiZr	9.00	1841	205	9	[35]
Al _{0.3} NbTaTi _{1.4} Zr _{1.3}	8.10	1965	243	4	[35]
Al _{0.3} NbTa _{0.8} Ti _{1.4} V _{0.2} Zr _{1.3}	7.60	1965	259	4	[35]
Al _{0.5} NbTa _{0.8} Ti _{1.5} V _{0.2} Zr	7.30	2035	279	3.5	[35]

yield strengths listed in Table 3. Fig. 9 shows the SYSs dependence of room temperature compressive SYSs for current alloys and other reported HEAs. It is seen that all the HEAs studied in present work possess relatively high SYS values, while their plastic strain vary in a wide range from about 10% to above 50%. Among these HEAs, M10V10 possesses the SYS of 251 KPa m³/kg and the plastic strain of 35%. MOV10 and M3V3 HEAs exhibit so excellent plasticity that they cannot be broken in the compression test (here the strains of MOV10 and M3V3 are noted as 50% in the following context). The refractory HEAs reported in literatures exhibit high SYSs, while their plastic strains are lower than 10%. The plastic strains for other HEAs composed of the late transition elements are lower than 35% except CrFeCoNiCu and AlCrFeCoNi HEAs. However, it is seen that the SYS of CrFeCoNiCu HEA is only 28 KPa m³/kg. Therefore, it can be concluded that the bcc type of HEAs developed so far still have insufficient plasticity at room temperature, or exhibit low strength. On the contrary, M3V3 developed in present work not only possesses plastic strain up to 50% but also has the SYS as high as 198 MPa m³/kg. In other words, M3V3 may have excellent ductility, which makes this kind of HEAs more advantageous than CrNbTiVZr HEAs for engineering application.

It is widely accepted that the solid solution strengthening of metallic solid solutions arises from the elastic interactions between the local stress field of solute atoms and dislocations [24,25]. The interaction force, f_m , increases with an increase of the atomic size difference and moduli misfit, δ_G , between the solute and solvent atoms. Here, we propose a shear modulus

mismatch, ΔG , to estimate the solid solution strengthening effect in HEAs. This parameter is expressed as $\Delta G = 100 \sqrt{\sum_{i=1}^n c_i \left(1 - \frac{G_i}{G_a}\right)^2}$, where c_i and G_i are the atomic percentage and shear modulus of the i th element, G_a the average shear modulus, respectively. Based on shear moduli of the components,

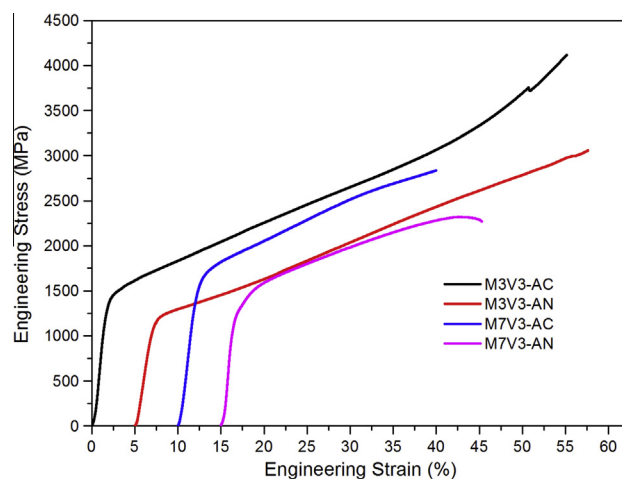


Fig. 8. The stress–strain curves of M3V3 and M7V3 in as-cast state and after annealed at 1273 K for 72 h.

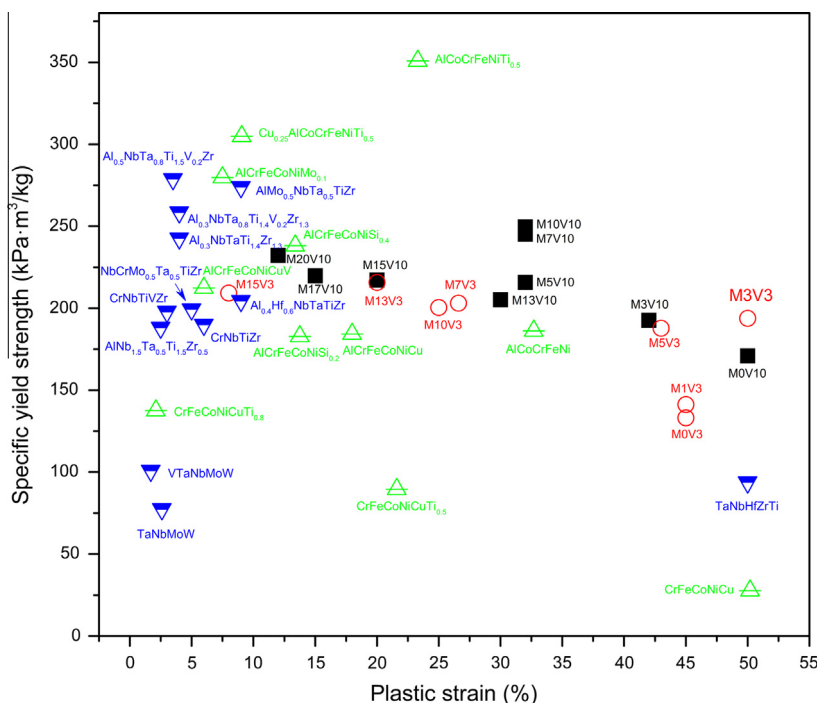


Fig. 9. The plot of the specific yield strength vs. plastic strain for the present HEAs and other reported HEAs composed of late transition and refractory elements.

the ΔG values of current HEAs were calculated, as shown in Table 3. It is seen that the addition of Mo into TiZrNbV alloys dramatically increases the ΔG of HEAs. Combined with the XRD analyses, it can be deduced that formation of single solid solution phase occurred when $\Delta G < 29$ in current alloy systems when prepared by copper mould casting. When ΔG is larger than 29, phase decomposition happened.

4. Conclusion

In this study, the microstructural evolution and compressive properties of a series of TiZrNbMoV HEAs were investigated. The alloying effects of Mo and V on the phase formation and strengthening effect of these HEAs were discussed. Based on the obtained results, conclusive remarks are drawn as following.

1. The phase formation is strongly affected by both the addition of Mo and V. For G-MV10 HEAs, there is a bcc type of structure as the Mo is lower than 25%, and two kinds of bcc phases as the Mo content is higher than 25%. Only one single bcc phase is formed in G-MV3 HEAs.
2. After annealed at 1273 K for 72 h, M3V3 still maintains single bcc phase, while the microstructure of M7V3 has been changed to two bcc phases and C15 type of Laves phases.
3. By using the existing criteria including δ , ΔH_{mix} , ΔS_{mix} , Ω and $\Delta\chi$, the phase composition were predicted for the present HEAs. It is found that all these parameters for the present HEAs fall in the region of solid solution phase proposed based on these criteria. It is also found that the G-MV3 HEAs are more liable to form solid solution microstructure than G-MV10 HEAs.
4. Mo plays a strong solid solution strengthening effect on this kind of HEAs. M3V3 demonstrates excellent comprehensive compressive mechanical properties with yield strength of 1312 MPa and plastic strain higher than 50%. The M3V3 and M7V3 HEAs maintain equivalent plastic deformation abilities to those of as-cast alloys after annealed at 1273 K for 72 h.

5. The TiZrNbMoV HEAs investigated in this work possesses relative high SYS among all the bcc HEAs. This strengthening effect is found to be correlated to the shear modulus mismatch. The solubility limit of current HEAs was found to be consist with $\Delta G = 29$, when prepared by copper mould casting.

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